higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PLUMBING
NQF LEVEL 2

12 MARCH 2019

This marking guideline consists of 5 pages.

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QUESTION 1: INTRODUCTION TO PLUMBING

1.1 Waste water is water that has been used and must be taken away from the building. (2)

1.2
- Solid
- Evaporation
- Liquid
- Transpiration
- Gas
- Condensation
- Precipitation (Any 3 × 1) (3)

1.3 Sewage screening (1)

1.4
- Planning for the job
- Briefing plumbers
- Waiting in readiness
- Overseeing the project (4)

1.5
- Bricklayer and concrete worker
- Plaster and tiler
- Paver
- Electrician
- Carpenter and joiner
- Telephone and communication technician (Any 5 × 1) (5)

[15]

QUESTION 2: PLUMBING PRINCIPLES AND CODE OF PRACTICE

2.1
- Lakes
- Rivers
- Dams (3)

2.2
- If the water is in a pipe it is on a downward slope so that the force of gravity pulls the water down.
- If it is forced through a pipe the pushing force is stronger than the pulling force of gravity. (2 × 2) (4)

2.3 Desalination plant (1)

2.4
- Joint acceptance scheme for water installation
- Agreement certificate
- SANS-tested system (3)

2.5 $P = D \times H \times g$ (2)

2.6 A water-reticulation system is a network of piping deep down under the streets varying in depth from 2 metres to 1 metre under street level. (2)
[15]

QUESTION 3: MATERIALS, TOOLS AND EQUIPMENT

3.1 3.1.1 Tin snip
3.1.2 Pipe cutter
(2 × 1) (2)

3.2

CAPILLARY FITTINGS	COMPRESSOR FITTINGS
• Made from copper • Solder	• Made from brass • O-ring/Compression ring

(2)

3.3 •Basins
•Baths
•Sinks
•Showers
• Bidets (Any 3 × 1) (3)

3.4 • Toilets
• Urinals
• Squatting pan (Any 2 × 1) (2)

3.5 • Independent scaffold
• Mobile scaffold (Any ONE) (1)
[10]

QUESTION 4: WATER-PIPE ASSEMBLY

4.1 To get rid of any rough burrs (1)

4.2 • Measure and cut the pipe.
• Ream the pipe.
• Select the tools and equipment needed to solder copper.
• Remove all dirt from the pipe.
• Put a thin cooling of flux onto the end of the pipe.
• Measure how much solder is needed and clamp the pipe into a vice on the workbench.
• Slot the T-piece (or another connector) over the pipe.
• Heat the copper pipe using a gas flame.
• Put the end of the solder wire at the joint.
• Clean the pipe. (10)

4.3 745 + 22 = 767 mm (1)

4.4
- Manually with the hand
- With an inside bending spring
- With an outside bending spring
- With a bending tool (Any 2 × 1) (2)

4.5
4.5.1 Modified polyvinyl chloride
4.5.2 Unplasticised polyvinyl chloride
4.5.3 High-density polyethylene
(3 × 1) (3)

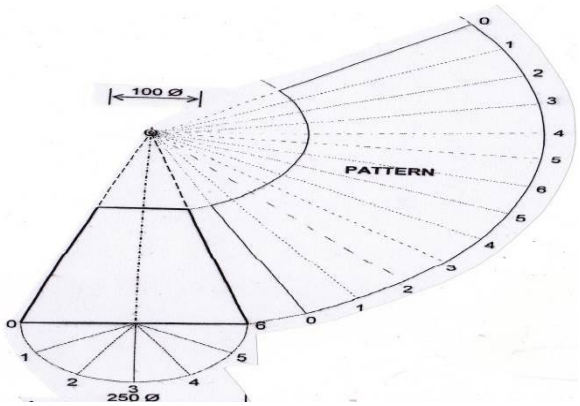
4.6
4.6.1 True
4.6.2 False
4.6.3 False
4.6.4 True
4.6.5 False
(5 × 1) (5)

4.7
- Glue called PVE – weld or solvent cement
- Rubber – O-ring couplings
- Electrofusion
- Brass compression fitting (Any 3 × 1) (3)

[25]

QUESTION 5: CUTTING AND JOINING OF METALS

5.1



Side view = 1
½ top view = 2
Pattern = 5

(8)

5.2
- Borax
- Silicon (2)

[10]

QUESTION 6: SET OUT BELOW-GROUND DRAINAGE SYSTEMS

6.1 So that the plumber can get into the pipes when necessary (2)

6.2 Trap (1)

6.3 6.3.1 Water that lies on top of the ground or building

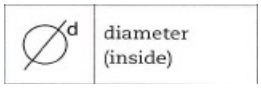
6.3.2 Dirty water used in a building that flows through pipes into the drainage system

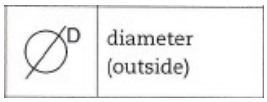
(2 × 2) (4)

6.4
- There should be enough pipes to lead the waste water to the system.
- The pipes must be as straight as possible between the places where they will be reached or accessed.
- Pipes must be properly supported so that they do not sag and cause drips in the middle or pull out of their joints.
- Pipes must be laid so that the water will flow easily down and through it.
- The joints on connections and the inspection chamber must be watertight.
- Drains should never run under a building. (6)

6.5 750 mm (1)

6.6 6.6.1 

6.6.2 

6.6.3 

(3 × 1) (3)

6.7
- Flexible transparent pipe
- Clean water
- Liquid detergent (dish-washing liquid)
- Two one-meter planks (slats)
- Marker to mark off on vertical surfaces (Any 3 × 1) (3)

6.8 Fall = distance × gradient (1)

6.9 It is to support the pipe all round so that it does not move or change direction. (2)

6.10 6.10.1 Lowest level of the inside of a pipe
6.10.2 Highest level of the outside of a pipe

(2 × 1) (2)

[25]

TOTAL: 100

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